Thus, $\sin\theta$ and cosec θ are positive and others are negative.
Similarly, we can find the sign of trigonometric functions in other two quadrants.
Let us illustrate the above discussions in Figure 3.10.

Signs of trigonometric functions in various quadrants can be remembered with the slogan

"All Students Take Chocolate" (ASTC rule)

Example 3.9 If $\sin\theta = \frac{3}{5}$ and the angle θ is in the second quadrant, then find the values of other five trigonometric functions.

Solution:

We know that $\sin^2\theta + \cos^2\theta = 1 \Rightarrow \cos\theta = \pm\sqrt{1-\sin^2\theta} = \pm\sqrt{1-\frac{9}{25}} = \pm\frac{4}{5}$

Thus, $\cos\theta = -\frac{4}{5}$ as $\cos\theta$ is negative in the second quadrant.

$$\sin\theta = \frac{3}{5} \Rightarrow \text{cosec}\,\theta = \frac{5}{3}; \quad \cos\theta = -\frac{4}{5} \Rightarrow \sec\theta = -\frac{5}{4}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = -\frac{3}{4}; \quad \cot\theta = -\frac{4}{3}$$

If $\sin\theta$ and $\cos\theta$ are known, then the reciprocal identities and quotient identities can be used to find the other four trigonometric values. The Pythagorean identities can be used to find trigonometric values when one trigonometric value and the quadrant are known.

3.4.3 Allied Angles

Two angles are said to be allied if their sum or difference is a multiple of $\frac{\pi}{2}$ radians.

Thus, any two angles of θ such as, $\pm\theta, \frac{\pi}{2}\pm\theta, \pi\pm\theta, \frac{3\pi}{2}\pm\theta, \ldots$, are all allied angles .

Now we shall find the trigonometric ratios involving allied angles θ and $-\theta$.

Trigonometric ratios of $-\theta$ in terms of θ

Let $\angle AOL = \theta$ and $\angle AOM = -\theta$. Let $P(a,b)$ be a point on OL. Choose a point P' on OM such that $OP = OP'$.

Draw PN perpendicular to OA intersecting OM at P'. Since $\angle AOP = \angle AOP'$ and $\angle PON = \angle P'ON$ $\triangle PON$ and $\triangle P'ON$ are congruent.

Thus, $PN = P'N$ and hence the point P' is given by $P'(a,-b)$

Now, by the definition of trigonometric functions

$$\sin\theta = \frac{b}{OP},\ \cos\theta = \frac{a}{OP},\ \tan\theta = \frac{b}{a}$$

Thus, $\sin(-\theta) = \frac{-b}{OP'} = \frac{-b}{OP} = -\sin\theta$

and $\cos(-\theta) = \frac{a}{OP'} = \frac{a}{OP} = \cos\theta$

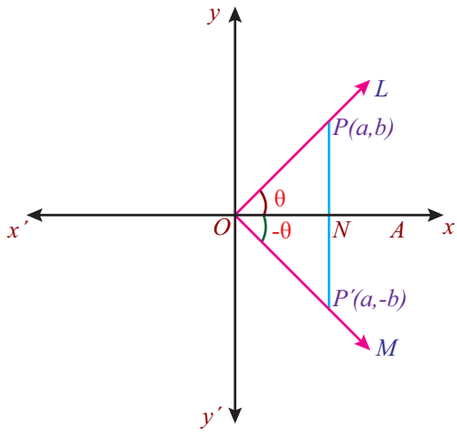


Figure 3.11